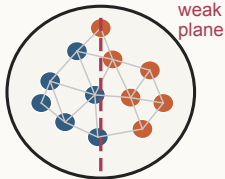


Intact bonded template

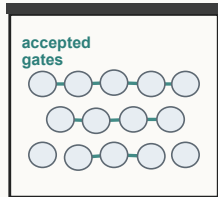
zero seating damage



1 mm Li₄SiO₄ mother pebble
500 bonded subparticles

Accepted bed load path

native force graph verified

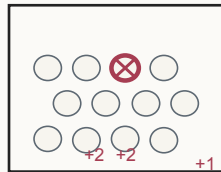


spanning load path
force-balance gate passed

Localized fracture chronology

one upper-bed pebble

5 of 493,500 bonds break



25 35 60

top displacement (um)

A hidden ceramic-breeder damage event becomes an auditable bond-loss and force-network history.